SUMMARY

- A solution is a homogeneous mixture of two or more chemically non-reacting substances mixed uniformly. The proportion of component which is more in the solution is called solvent and the lesser component is called solute.
- The different concentration units used to prepare the solutions are formality, molality , normality, molarity, molefraction, %w/w solution, %w/v solution, %v/v solution. If the solute quantity is very minimum then ppm unit is used to express its concentration.
- The standard solutions are prepared and diluted to desired concentration (working standards). This helps in overcoming the error due to weighing and maintain efficiency and consistency in the preparation of solution.
- Solubility is defined as the amount of solute in a saturated solution per 100g of a solvent. Solubility of a solute in a solvent depends on the nature of the solute and solvent , temperature.
- The solubility of a gas in a liquid depends upon, the nature of the gas and the nature of the liquid, the temperature of the system, and the pressure of the gas.
- The effect of pressure on the solubility of a gas in a liquid is governed by Henry's Law. It states that the solubility of a gas in a liquid at a given temperature in directly proportional to the partial pressure of the gas.
- The vapour pressure of a liquid is the pressure exerted by its vapour when it is in dynamic equilibrium with its liquid, in a closed container. According to Raoults Law, the vapour pressure of a solution containing a non-volatile solute is directly proportional to the mole fraction of the solvent (X_A). The proportionality constant being the vapour pressure of the pure solvent.
- The solution shows positive deviation from Raoult's Law if its vapour pressure is higher than that predicted by Raoult's Law for example ethyl alcohol and cyclohexane.
- The solution shows negative deviation if its vapour pressure is lower than that predicted by Raoult's Law for example acetone and chloroform.
- Colligative properties of solutions are those properties which depend only upon the number of solute particles in the solution and not on their nature. Such properties are
- (a) Relative lowering in vapour pressure: On addition of a non volatile solute it is observed that the vapour pressure of the solution is lesser than the solvent. According to Raolut's law relative lowering of vapour pressure is equal to the mole fraction of the solute.
- (b) Elevation of boiling pointΔT_b: The solution boils at a higher temperature than the pure solvent. This is due to the decreased vapour pressure of solution and the temperature it reaches the atmospheric pressure on heating to boil is much higher than the solvent.
- (c) Depression of freezing point:

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